

ELISA (Enzyme-linked Immuno Sorbent Assay): It all starts with the RIA (Radio Immuno Assay) and insulinemia

17 1950-1954 — the starting point

Yalow (medical physicist) and Berson (physician) collaborate at the Bronx Veterans Hospital.

Clinical problem: some diabetics treated with animal insulin become resistant to treatment.



Rosalyn Yalow

At Bronx Veterans Administration in 1977, just after she had won the Nobel Prize in Physiology or Medicine for the RIA.

R. Yalow was especially interested in insulin because her husband was diabetic.



Solomon Berson

He died five years before the Nobel Prize for the RIA was given.

Insulin Clearance: Normal

Characteristic	Normal Insulin Clearance
Main elimination organs	Liver (50–60%), kidneys (30–40%), peripheral tissues
Plasma half-life	Very short: 4–6 minutes
Circulating form	Free insulin → rapidly taken up and degraded
Disappearance curve	Simple, rapid exponential decay

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Clinical problem: some diabetics treated with animal insulin become resistant to treatment.

Hypothesis: injected insulin triggers an immune response

→ production of anti-insulin antibodies.



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Insulin Clearance: Normal vs With Anti-Insulin Antibodies

Characteristic	Normal Insulin Clearance	Clearance with Anti-Insulin Antibodies
Main elimination organs	Liver (50–60%), kidneys (30–40%), peripheral tissues	Liver, kidneys, tissues, but slowed by antibody binding
Plasma half-life	Very short: 4–6 minutes	Prolonged (insulin protected by antibodies)
Circulating form	Free insulin → rapidly taken up and degraded	Antibody-bound insulin → stable immune complexes
Disappearance curve	Simple, rapid exponential decay	Slower decay, sometimes biphasic (progressive release of bound insulin)

ELISA (Enzyme-linked Immuno Sorbent Assay): It all starts with the RIA (Radio Immuno Assay) and insulinemia

1955-1956 — Experiments with radioactive insulin

Insulin labeled with ^{131}I .

Injection into volunteers and measurement of the rate of insulin disappearance from the blood.

Result: in patients treated with insulin, the hormone binds to serum proteins → indirect evidence of anti-insulin antibodies.



1957-1959 — Birth of the RIA principle

Observation: if radioactive insulin is mixed with serum containing antibodies, some of it is retained, while the rest remains free.

Brilliant idea: if non-radioactive insulin is added, it “competes” to bind to the antibodies → the more there is, the less radioactive insulin is retained.

This competitive phenomenon makes it possible to quantify insulin in unknown samples.

1960 — Formalization of the radioimmunoassay

Yalow and Berson publish the complete method.

RIA becomes the most sensitive tool of the time for measuring hormones, vitamins, viruses, drugs, etc.

1977 — Nobel Prize in Physiology or Medicine

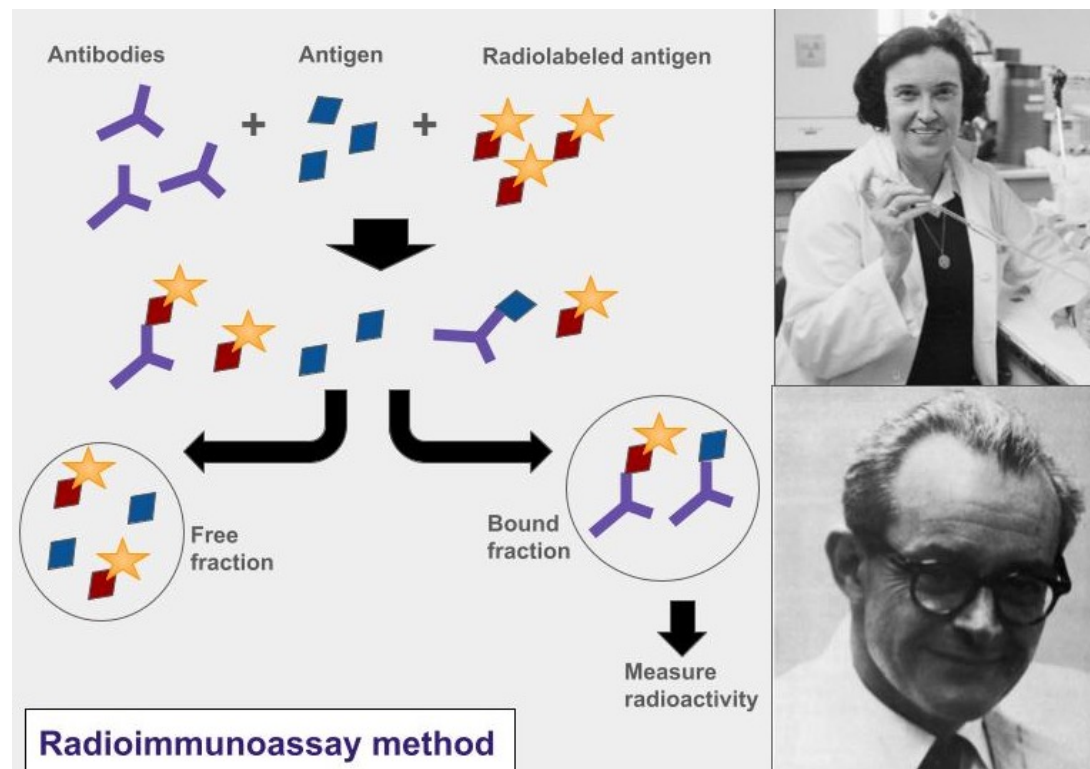
Rosalyn Yalow receives the prize (Berson died in 1972 and could not be awarded the prize).

ELISA (Enzyme-linked Immuno Sorbent Assay): It all starts with the RIA (Radio Immuno Assay) and insulinemia

Hormones concentration in the blood: 10^{-9} et 10^{-12} M

Antibody affinity for an antigen (K_d) 10^{-9} à 10^{-12} M

Begining of the Radio Immuno Assay (RIA)



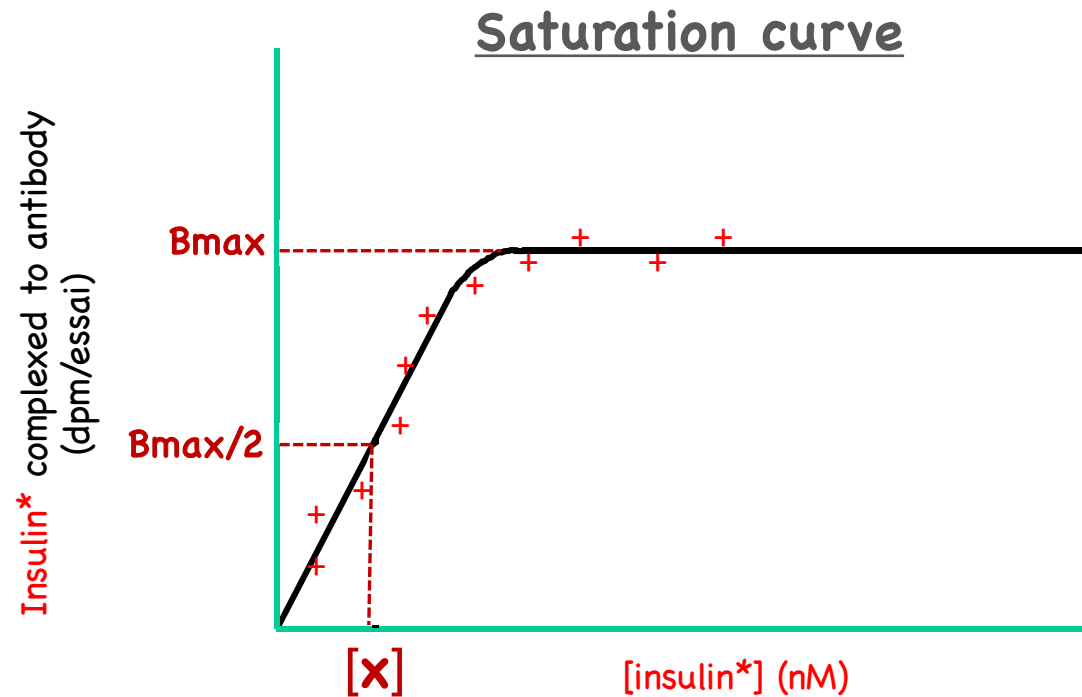
Determining the insulin concentration of a blood sample:

Principle of the saturation curve

1. Mix known increasing amounts of radioisotope-tagged insulin (insulin*) and a fixed (A) concentration of antibodies to form an insulin-antibody complex
2. Incubate long enough for the insulin* to bind with antibodies and reach the binding equilibrium status
3. Following incubation, the complexes are isolated from free antibodies and free insulin* by ultra-centrifugation
4. The radioactivity emitted by each pellet (increasing concentration of bound insulin*) is determined in a radiation counter
5. Establish the saturation curve.

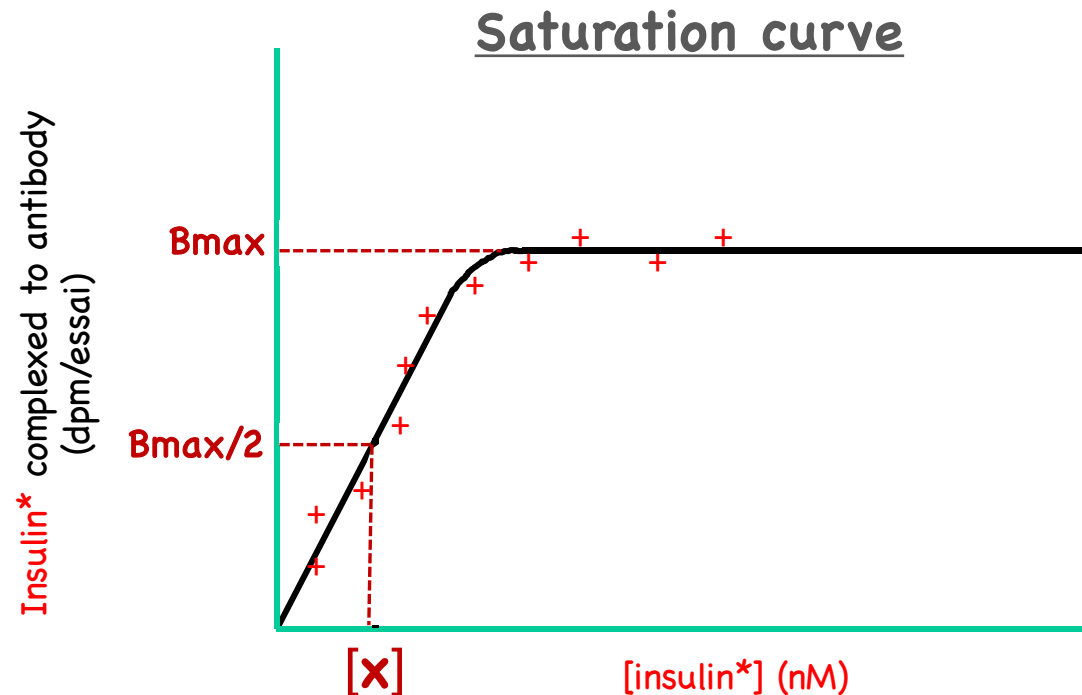
Determining the insulin concentration of a blood sample: *Principle of the saturation curve*

1. Mix known increasing amounts of radioisotope-tagged insulin (insulin^{*}) and a fixed (A) concentration of antibodies to form an insulin-antibody complex
2. Incubate long enough for the insulin^{*} to bind with antibodies and reach the binding equilibrium status
3. Following incubation, the complexes are isolated from free antibodies and free insulin^{*} by ultra-centrifugation
4. The radioactivity emitted by each pellet (increasing concentration of bound insulin^{*}) is determined in a radiation counter
5. Establish the saturation curve.
6. Determine the concentration of insulin^{*} [X] for **B_{max}/2**.



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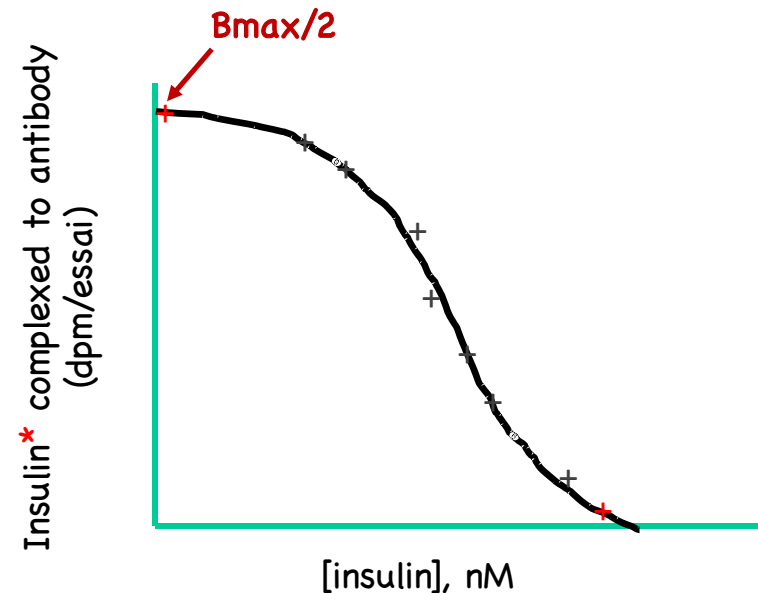


This will be used for the competition experiment meant to establishing the calibration curve and evaluation of the blood sample concentration of insulin

Determining the insulin concentration of a blood sample: *Principle of the competition experiment and calibration curve*

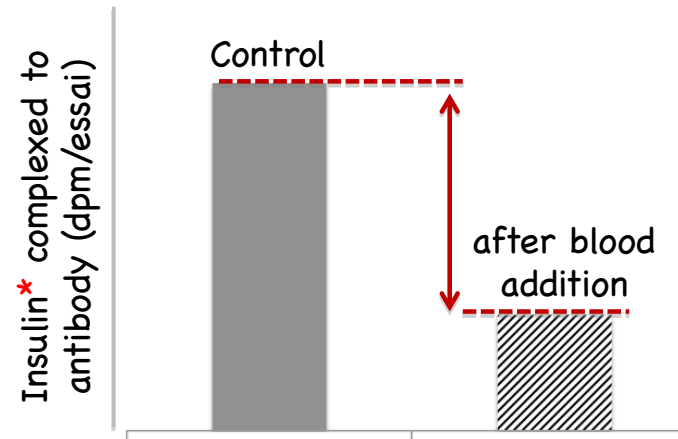
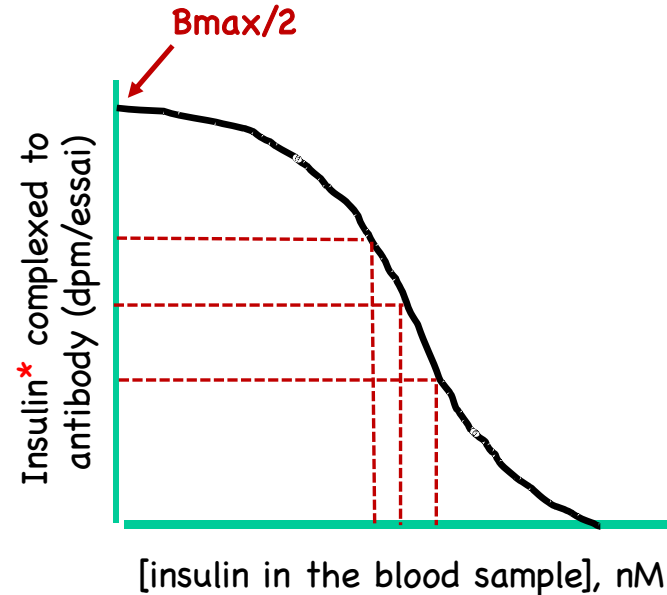
1. Use **X nM** of insulin* (determined previously for $B_{max}/2$) with the fixed (**A**) concentration of antibodies and increasing known concentrations of non radio-labelled insulin (insulin) and incubate the mixture to reach the binding equilibrium status calibration
2. Ultracentrifugation
3. Use of a radiation counter to measure the radioactivity emitted by the pellet.
4. Establish the calibration curve

Calibration curve



Determining the insulin concentration of a blood sample: *Principle of the competition experiment and calibration curve*

1. Perform the same competition experiment but with different amount of blood sample (or same volume from different dilutions) instead of increasing concentrations of non radio-labelled insulin
2. Plot the radioactivity of the bound fraction against the corresponding analyte (here, insulin) concentrations
3. Determine the concentration of the analyte (insulin) in the blood sample

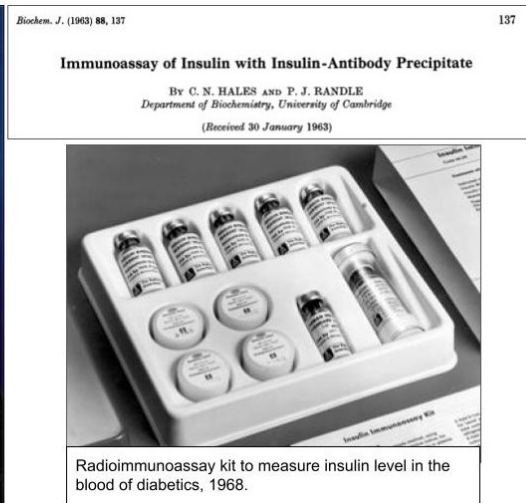


ELISA (Enzyme-linked Immuno Sorbent Assay) :

Assay Procedure Flow Chart

Day One						Day Two		
Set-up	Step 1	Steps 2 - 3	Step 4	Step 5	Step 6	Step 7	Step 8	Steps 9 - 11
Tube Number	Add Assay Buffer	Add Standards/ QC/ Samples	Add ¹²⁵ I- Insulin Tracer	Add Insulin Antibody	Vortex, Cover, and Incubate 20-24 hrs at 4°C	Add Precipitating Reagent	Vortex, and Incubate 20 min. at 4°C	Centrifuge @ 4°C for 20 min., Decant and Count
1,2	-	-	100 µL	-		-		
3,4	200 µL	-	100 µL	-		1.0 mL		
5,6	100 µL	-	100 µL	100 µL		1.0 mL		
7,8	-	100 µL of 0.156 ng/mL	100 µL	100 µL		1.0 mL		
9,10	-	100 µL of 0.313 ng/mL	100 µL	100 µL		1.0 mL		
11,12	-	100 µL of 0.625 ng/mL	100 µL	100 µL		1.0 mL		
13,14	-	100 µL of 1.25 ng/mL	100 µL	100 µL		1.0 mL		
15,16	-	100 µL of 2.5 ng/mL	100 µL	100 µL		1.0 mL		
17,18	-	100 µL of 5.0 ng/mL	100 µL	100 µL		1.0 mL		
19,20	-	100 µL of 10.0 ng/mL	100 µL	100 µL		1.0 mL		
21,22	-	100 µL of QC 1	100 µL	100 µL		1.0 mL		
23,24	-	100 µL of QC 2	100 µL	100 µL		1.0 mL		
25,26	-	100 µL of unknown	100 µL	100 µL		1.0 mL		
27-n	-	100 µL of unknown	100 µL	100 µL		1.0 mL		

ELISA (Enzyme-linked Immuno Sorbent Assay): from RIA to ELISA



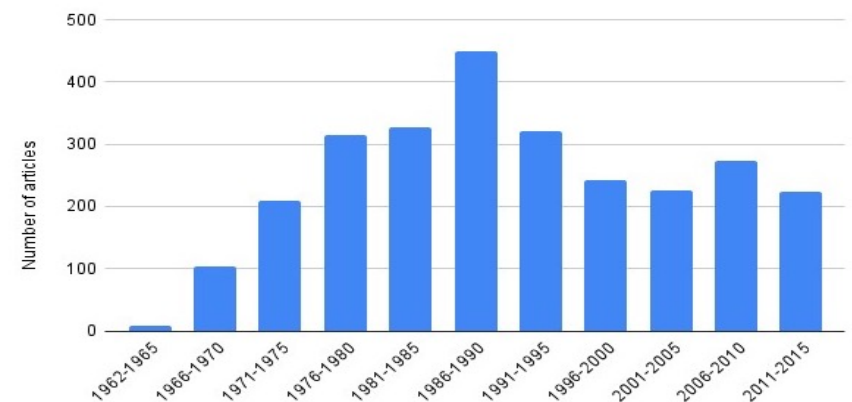
Hale's investigation of diabetes was prompted by the death of his paternal grandfather due to the disease, and by the fact that both his mother and sister also had diabetes.

Hales's goal was to improve the RIA, making it simpler and faster to perform, as well as more sensitive and specific.

RIA had a profound impact in clinical medicine :

- it provided means to diagnose conditions caused by small changes in the quantity of hormones that could not be detected by previous methods.
- able to be used for a wide variety of biological substances, RIA opened the way to new insights in many different areas (endocrinology, immunology, cardiology, gastroenterology, nephrology and neuroscience).

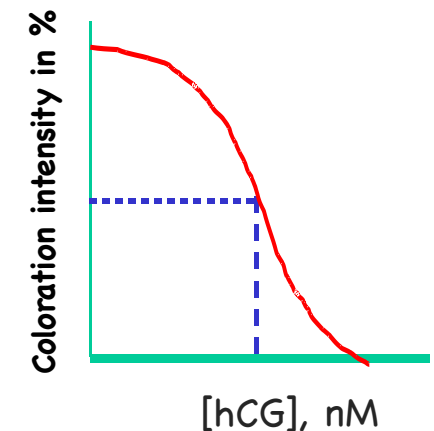
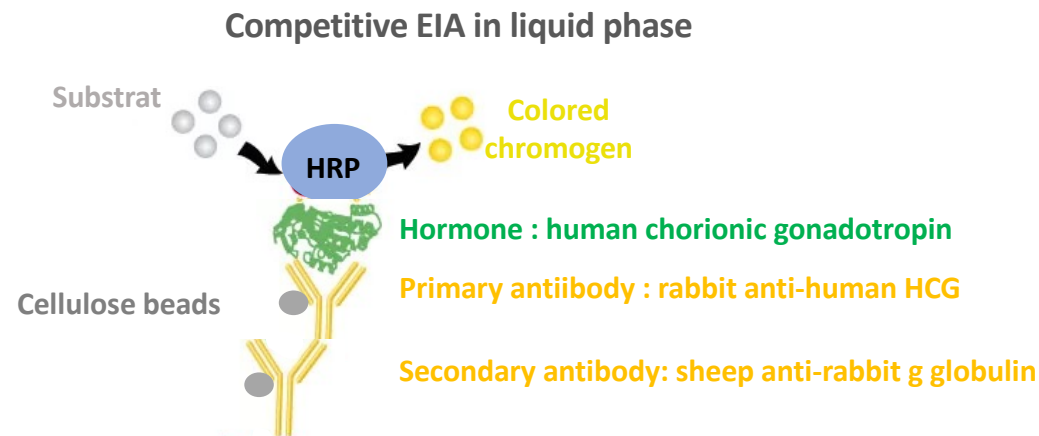
Number of articles with the word 'RIA' in their titles (1962-2015)



ELISA (Enzyme-linked Immuno Sorbent Assay) : from RIA to ELISA through EIA

1. the RIA was a *technically demanding procedure*
2. Radioactive labels had a *limited shelf-life*, were *costly*, and made the testing process difficult to *automate*
3. *Radiation exposure* for laboratory staff and their disposal as waste became unacceptable.

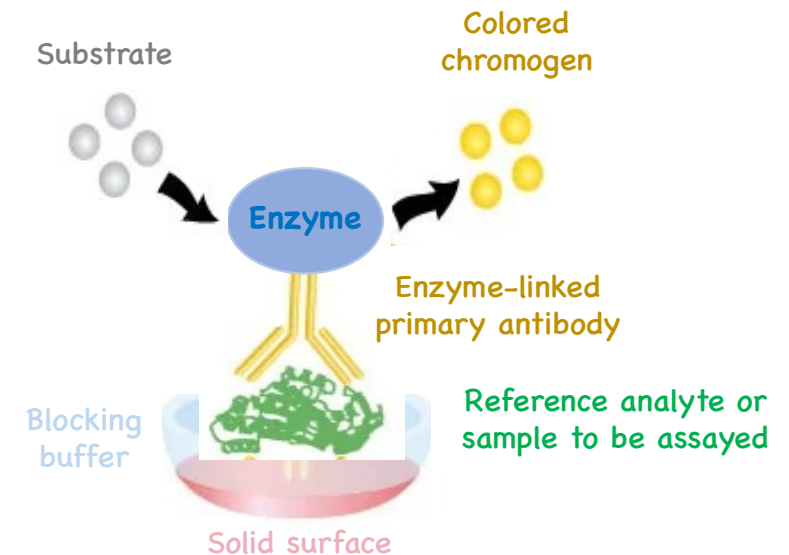
Engvall, Perlmann, Schurrs and Van Weeman set out to develop a test based on **enzymatic activity instead of radioactivity**. They conjugated **Horseradish Peroxidase (HRP)** with human chorionic gonadotropin (hCG). This method made it possible to detect small quantities of the hormone, 6–8 days after the fertilised egg gets implanted into the uterus



Principle of direct ELISA

ELISA, like EIA, uses the catalytic properties of enzymes to quantify the number of analyte/Ab complexes.

- Adsorbing the reference analyte (hormone) non-specifically or covalently to the surface of a solid phase (96-well plate well, magnetic particle or plastic bead)
- Blocking: with BSA, ovalbumin, aprotinin, or other animal proteins
- Adding the antibody covalently coupled to an enzyme
- Separating free from complexed, Washing (PBS) the complexed
- Adding a chromogenic substrate (oxidation of H_2O_2 by horseradish peroxidase, or hydrolysis of p-nitrophenyl-phosphate by alkaline phosphatase)
- Reading the emission of a measurable colored chromogen (blue for HRP, yellow for AP)



Positive control:
Standard curve

Negative controls:
Ab without Ag
Ag without Ab

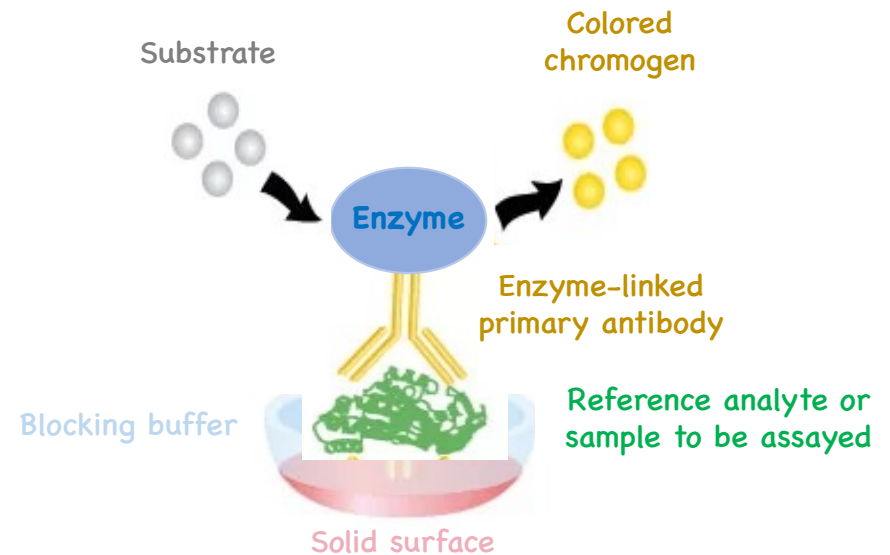
Principle of direct ELISA

ELISA, like EIA, uses the catalytic properties of enzymes to quantify the number of analyte/Ab complexes.

- Adsorbing the reference analyte/sample
- Blocking
- Addition of enzyme-linked Ab I
- Wash step
- Incubation with substrate and measure

Positive control:
Standard curve

Negative controls:
Ab without Ag
Ag without Ab



Advantages: Rapid due to few steps

Disadvantages: Low sensitivity (no signal amplification), high cost (conjugation of a specific antibody is required for each analyte), high background interference (non-specific antigen immobilization)

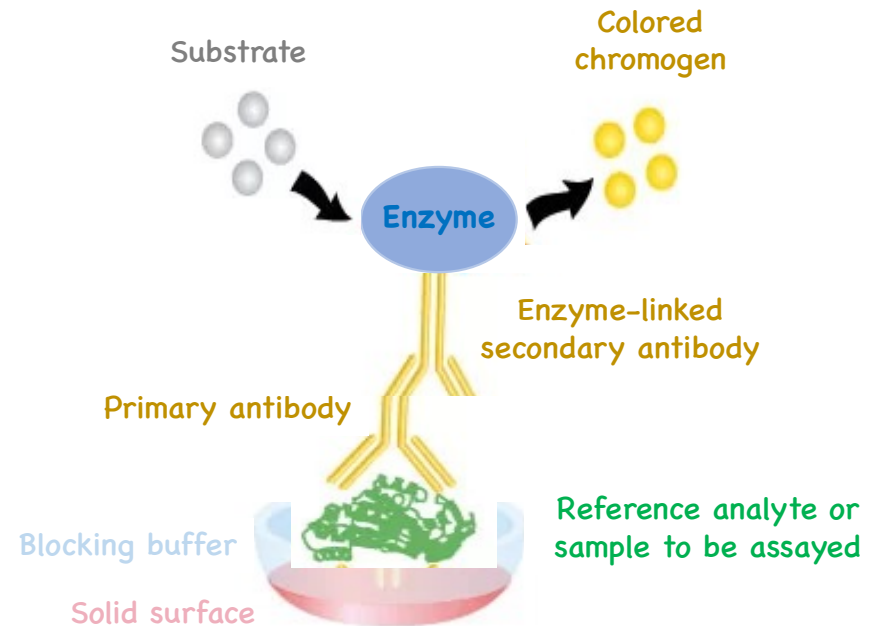
Principle of indirect ELISA

Indirect ELISA requires two antibodies: a primary detection antibody that binds to the analyte of interest, and an enzyme-linked secondary antibody that binds to the primary one.

- Adsorbing the reference analyte/sample
- Wash step
- Addition of Ab I
- Wash step
- Addition of enzyme-linked Ab II (polyclonal)
- Incubation with substrate and measure

Positive control:
analyte + Ab I + II

Negative controls:
Ab I + II without Ag
Ab II + Ag without Ab I



Advantages: Higher sensitivity, less expensive, and more flexible (thanks to the numerous possible primary antibodies) than direct ELISA.

Disadvantages: Risk of cross-reactivity between primary and secondary detection antibodies.

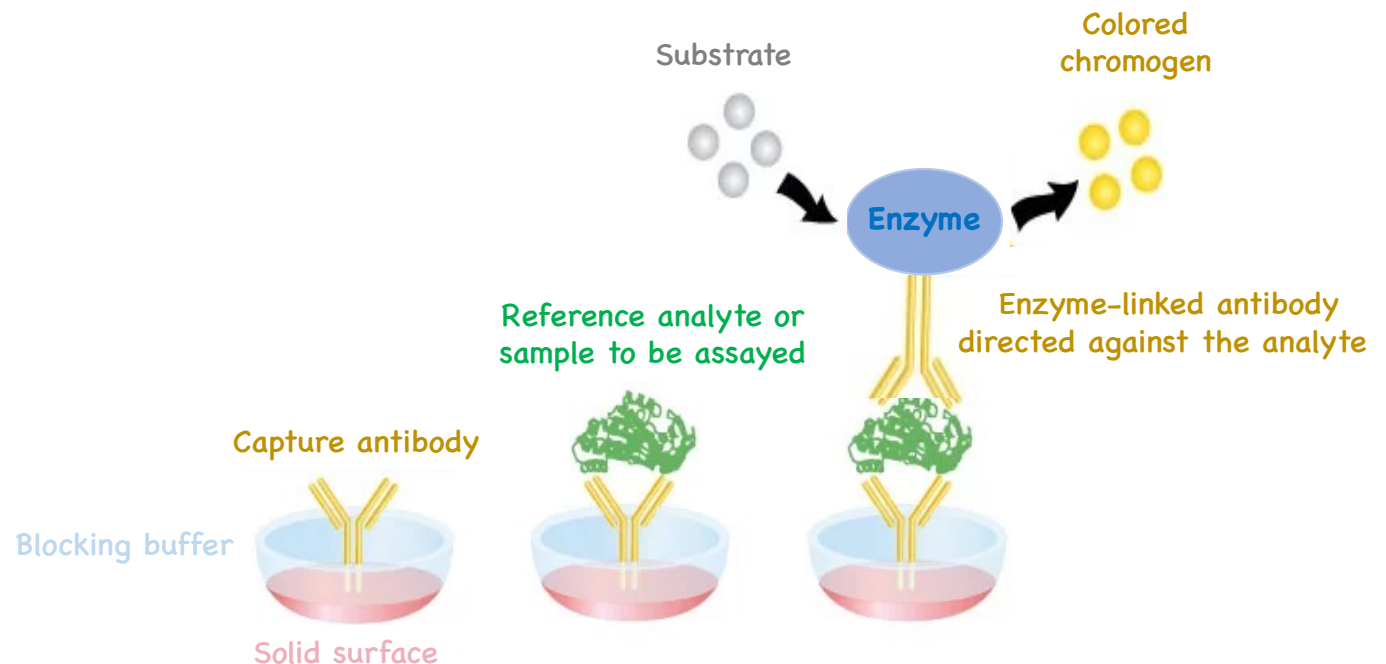
Principle of sandwich ELISA

Sandwich ELISA requires two antibodies: a primary detection antibody that binds to the analyte of interest, and an enzyme-linked secondary antibody that binds to the primary one.

- Addition of the reference analyte/sample
- Wash step
- Addition of enzyme-linked Ab
- Wash step
- Incubation with substrate and measure

Positive control:
analyte + Ab

Negative control:
Ab without Ag



Advantages: Sandwich ELISAs have the highest specificity of all ELISAs.

Disadvantages: Costly, and the need for a matched pair (divalent/multivalent analyte)

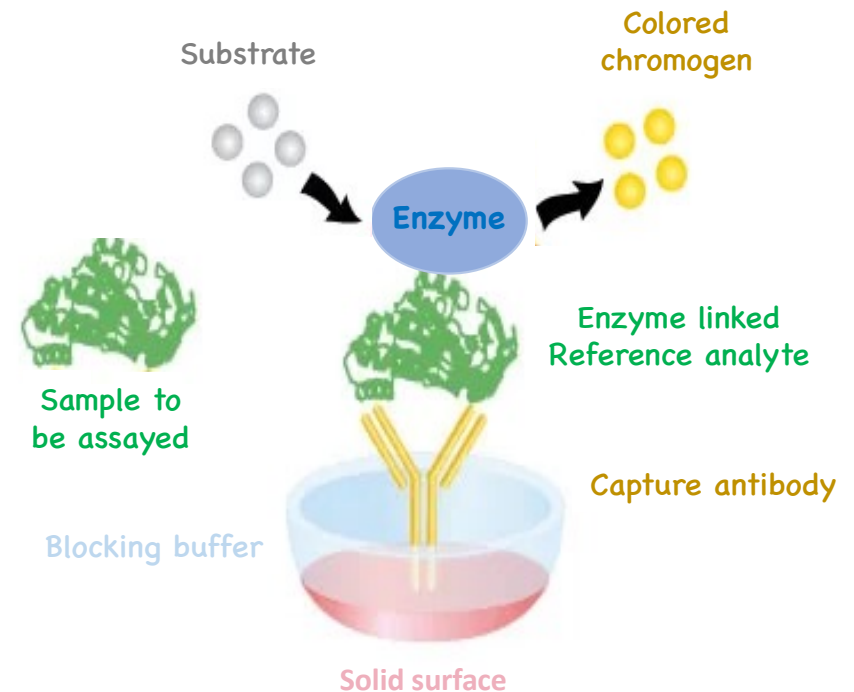
Principle of competitive ELISA

This type of ELISA uses two analytes, as previously described for RIA: a purified analyte conjugated to an enzyme, and the same unconjugated analyte present (or not) in the fluid to be tested.

- The combination of both conjugated and unconjugated hormones in the wells allows competition for antibody binding.
- A color indicates a negative test, as the antibodies are saturated by the conjugated hormone.
- The absence of color indicates a positive test and the presence of hormones in the serum to be tested

Positive control:
Unconjugated
analyte in excess

Negative control:
No sample



Advantages: Reduced sample purification, measurement of a wide range of analytes in a given sample, and usable for small analytes (compared to sandwich ELISA).

Disadvantages: Cannot be used on diluted samples

Applications of Elisa in diagnostic tests

- **Detection and quantification of hormone expression levels:**

Luteinizing hormone (LH), Follicular stimulating hormone (FSH), Prolactin (PL), Testosterone, Human chorionic gonadotropin (hCG), insulin.

- **Detection and quantification of the presence of blood antibodies:**

Antibodies linked to bacterial (Lyme disease, brucellosis, syphilis), viral (Hepatitis A, B, C, HIV) or fungal () infectious diseases.

- **Screening of viral contaminants in blood/plasma donation samples:**

Anti-HIV-1/2, Anti-HCV, HBsAg.

- **Tracking Disease Outbreaks**

Cholera, Influenza

- **Detection and quantification of expression levels of serum tumor markers:**

Prostate-specific antigen (PSA), Carcinoembryonic Antigen (CEA)

- **Drug detection:**

Amphetamine and derivatives (methamphetamine and 3,4-methylenedioxymethamphetamine), cocaine, benzoylecgonine

- **Detection of 2nd messengers in research:**

AMPc, GMPc

- **Use in food safety tests:**

Detection of mycotoxins and allergens, gluten

Factors that can interfere with appropriate ELISA test can occur at any phase of the process, beginning with specimen collection

Plate Assay: shape and quality of the wells, material of the plate, potential pre-activation, coating

Buffer: pH, contamination

Capture and detection antibody: incubation time, temperature, specificity, titer, affinity

Blocking buffer: cross-reactivity, concentration, contamination

Target analyte: conformation, stability, epitopes.

Enzyme conjugate: type, concentration, function, cross-reactivity

Washes: contamination, frequency, volume, duration, composition

Substrate: quality/manufacturer

Detection: instrument-dependent factors

Reader/human error

Exercise 1

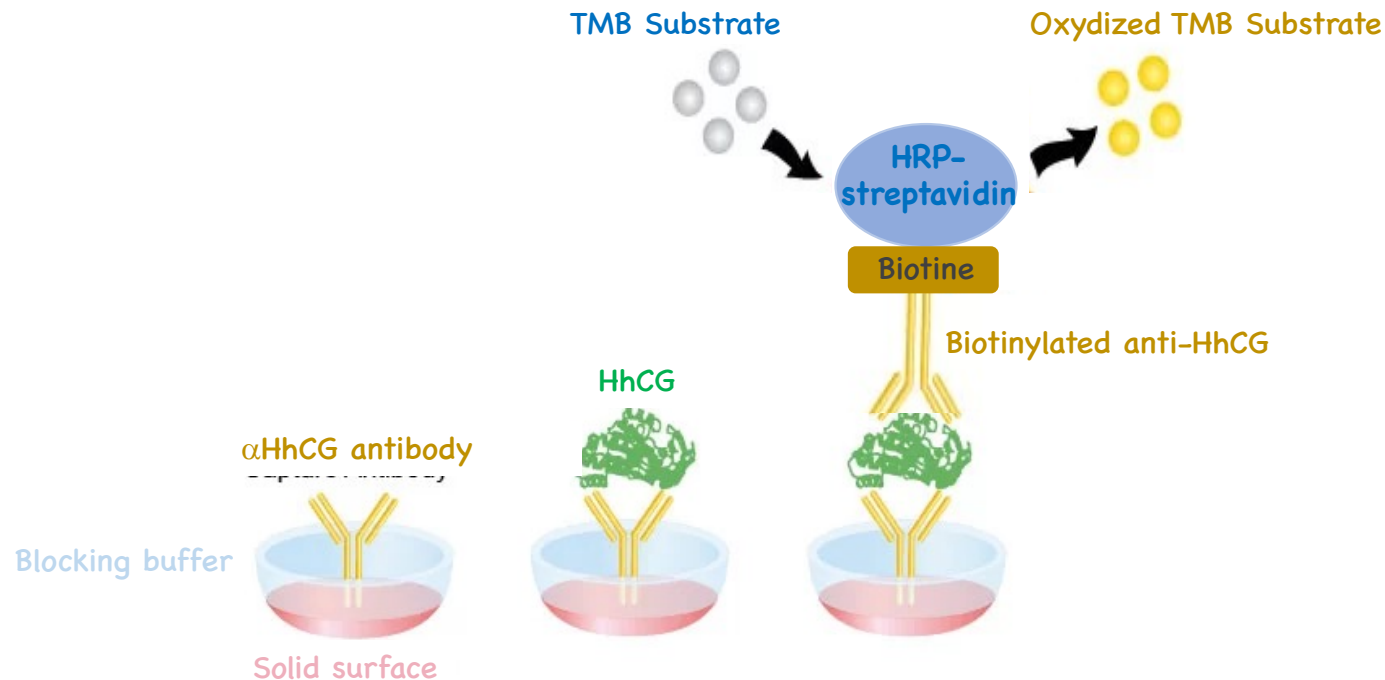
« The Human hCG (HhCG) (human Chorionic Gonadotropin) ELISA kit is an *in vitro* enzyme-linked immunosorbent assay for the quantitative measurement of human hCG in serum, plasma, and cell culture supernatants. »

« This assay employs an antibody specific for HhCG coated on a 96-well plate. Standards and samples are pipetted into the wells and hCG present in a sample is bound to the wells by the immobilized antibody. The wells are washed and biotinylated anti-human hCG antibody is added. After washing away unbound biotinylated antibody, HRP-conjugated streptavidin is pipetted to the wells. The wells are again washed, a TMB substrate solution is added to the wells, and color develops in proportion to the amount of hCG bound. The Stop Solution changes the color from blue to yellow, and the intensity of the color is measured at 450 nm. »

1. What type of ELISA does this kit use ? Make a scheme to explain it.

Exercise 1

This assay employs an antibody specific for HhCG coated on a 96-well plate. Standards and samples are pipetted into the wells and hCG present in a sample is bound to the wells by the immobilized antibody. The wells are washed, and biotinylated anti-human hCG antibody is added. After washing away unbound biotinylated antibody, HRP-conjugated streptavidin is pipetted to the wells. The wells are again washed, and a TMB substrate solution is added to the wells. The wells are again washed, and a TMB substrate solution is added to the wells.



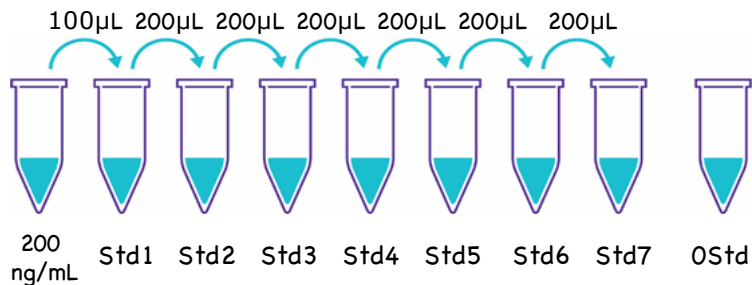
Exercise 1

1. Preparation of standard: Briefly spin a vial of Standard Protein. Add 400 μL 1X Assay Diluent into the Standard Protein vial to prepare a 200 ng/mL standard HhCG. Dissolve the powder thoroughly by a gentle mix. Add 100 μL hCG standard (200 ng/mL) from the vial of Standard Protein, into a tube with 400 μL Assay Diluent to prepare a 40,000 pg/mL standard solution. Use the 40,000 pg/mL standard solution to produce a dilution series (shown below). Mix each tube thoroughly before the next transfer. Assay Diluent serves as the zero standard (0 pg/mL).

2. Sample dilution: 1X Assay Diluent should be used for dilution of standard, serum and plasma samples. The suggested dilution for normal serum/plasma is 2 – 10-fold.

Note: Levels of hCG may vary between different samples. Optimal dilution factors for each sample must be determined by the investigator.

Preparation of standard:



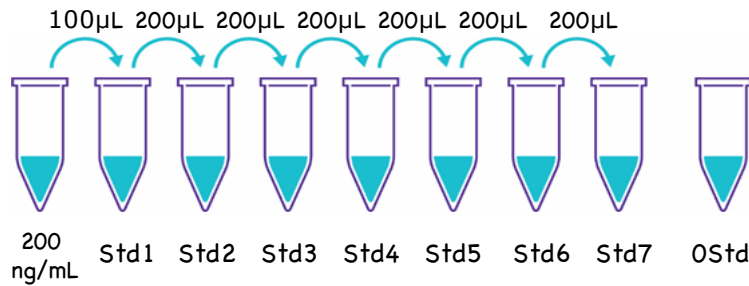
		Std1	Std2	Std3	Std4	Std5	Std6	Std7	0Std
Diluent Volume	Std+ 400 μL	400 μL	400 μL	400 μL	400 μL	400 μL	400 μL	400 μL	400 μL
Concentration	200 ng/mL	40000 pg/mL	13333 pg/mL	4444 pg/mL	1481 pg/mL	493,4 pg/mL	164,6 pg/mL	54,87 pg/mL	0 pg/mL

Exercise 1

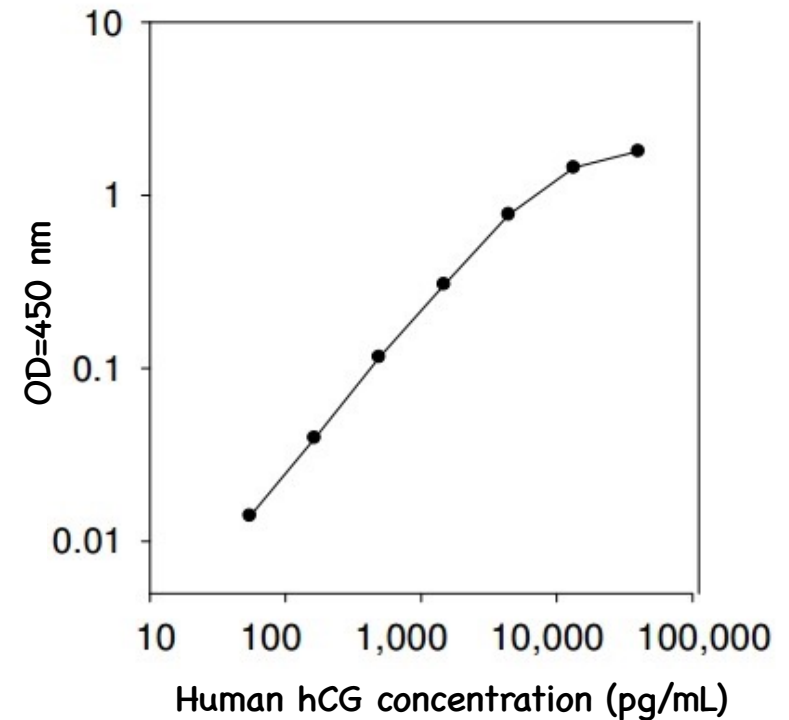
Assay Procedure

1. Bring all reagents and samples to room temperature (18–25 °C) before use. It is recommended that all standards and samples be run at least in duplicate.
2. Add 100 µL of each standard and sample into appropriate wells. Cover wells and incubate for 2.5 hours at room temperature with gentle shaking.
3. Discard the solution and wash 4 times with 1X Wash Solution.
4. Add 100 µL of 1X prepared biotinylated antibody to each well. Incubate for 1 hour at room temperature with gentle shaking.
5. Discard the solution. Repeat the wash as in step 3.
6. Add 100 µL of prepared Streptavidin solution to each well. Incubate for 45 minutes at room temperature with gentle shaking.
7. Discard the solution. Repeat the wash as in step 3.
8. Add 100 µL of TMB One-Step Substrate Reagent to each well. Incubate for 30 minutes at room temperature in the dark with gentle shaking.
9. Add 50 µL of Stop Solution to each well. Read at 450 nm immediately.

Exercise 1



		Std1	Std2	Std3	Std4	Std5	Std6	Std7	OStd
Diluent Volume	Std+ 400µL	400µL	400µL	400µL	400µL	400µL	400µL	400µL	400µL
Concentration	200 ng/mL	40000 pg/mL	13333 pg/mL	4444 pg/mL	1481 pg/mL	493,4 pg/mL	164,6 pg/mL	54,87 pg/mL	0 pg/mL



2. Why is it important to make serial dilutions ?
3. Why is it « recommended that all standards and samples be run at least in duplicate. » ?
3. What is the measurable concentration range for the sample ?
4. How can you explain the end of the curve shape ? What does it means ?